[Experiment, Analysis, and Benchmark] Private Supervision under Strong Public Representations: Label Access, Release Granularity, and Verification

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***Abstract*—Strong public representations make task supervision the scarce resource of a data-platform classifier: public labels under some acquisition contract, or private client opinions reachable only through aggregate-only, differentially private interfaces. This experiment and analysis paper measures what private supervision adds and what verification costs, fixing annotation budget and acquisition mode, whole-client privacy budget, report payload, representation and public post-processing on four tasks; image clients are simulated privacy units built from public benchmarks, Usenet authors and FEMNIST writers are natural. Three findings stand out. First, acquisition rights decide whether private reports are needed: with ten random CIFAR labels at $\varepsilon = .1$, private regional naming adds 13.0 points over the strongest of fourteen public learners, yet ten feature-directed labels reach the same 91.6% classifier without private reports. Second, raw sensitivity does not rank release interfaces: per-example votes are about 14 times more sensitive than regional reports, but pooled within regions they carry .88–1.07 times the regional noise and match regional accuracy within 2.3 points; payload is what differs. Third, publication evidence is a separate resource: 12.7% of the classifiers a mean-gain certificate publishes recall some class below 10%, and a budget split selected on independent development cohorts and frozen raises all-class acceptance on new cohorts from 5.4% to 78.2% at $\varepsilon = 1$, with no false accept. The public artifact (https://github.com/Xinbw/SDD) regenerates every reported number.**

***Index Terms*—differential privacy, experimental analysis, federated data, distributed DP, data acquisition, public representations, private verification**

I. Introduction

Strong public representations have changed where the cost of a classifier lies. A frozen vision or text encoder already places semantically similar records near each other, so a data platform can expose class structure before it owns a single task label. What remains scarce is *supervision*: public annotation, at a cost and under acquisition rights that determine which examples may be requested, or private clients, such as devices, organizations or user accounts, whose labelled records are reachable only through protected aggregate interfaces.

A platform that wants to publish a classifier therefore makes three decisions (Figure 1): *what supervision to buy*, public labels or private reports; *what statistic to release* from clients, per-example votes, a compact regional summary or a clipped model head; and *what evidence to require* before publication. Evaluated separately and judged by average accuracy, reported gains mix these resources and do not transfer.



Fig. 1. Three platform decisions under a shared resource contract: the acquisition contract for public labels and private supervision, the protected release object, and the evidence required before publication.

A. An experiment and analysis study

This paper is an experimental analysis, not a new release mechanism: regional naming, distributed discrete-Gaussian aggregation and noisy-count confidence bounds are established components. Its contribution is a *resource-matched evaluation protocol* and what it reveals. The protocol measures private value against the strongest public learner with the same labels, representation and acquisition mode (Section II); every comparison fixes annotation budget and acquisition mode, total whole-client privacy budget, released coordinates and public post-processing, and verification is charged to the same total budget as naming. We apply it to 3426 inherited primary records over 90 cohorts and 4224 public label sets, and to five

studies whose designs were frozen before they ran (Table III). Image clients are simulated privacy units built from public benchmarks; Usenet authors and FEMNIST writers are natural.

B. Findings

a) Finding 1: acquisition rights decide private value: With ten random CIFAR-1k labels at $(\varepsilon, \delta) = (.1, 10^{-5})$, joint naming beats the strongest of fourteen equal-budget public learners by 13.0 points on average (median 13.6), though 9% of label draws lose. Ten feature-directed labels recover the same 91.6% classifier without private reports: an evaluation must fix both how many labels exist and how they are acquired. With a weaker ResNet-18 encoder, directed labels again reach the partition ceiling without private reports, but random-label joint naming changes the strongest public learner by -4.0 points: passive private value needs regions carrying the classes.

b) Finding 2: raw sensitivity does not rank release interfaces: Per-example votes on $|P| = 2000$ proxy points are $\sqrt{|P|/K} \approx 14$ times more sensitive than a $K = 10$ regional report and need 20000 rather than 100 coordinates. Yet averaging point votes inside each region brings their noise to .88–1.07 times the regional level, and pooled point and regional joint naming differ by at most 2.3 points on all 8 tested settings. The protected head, a different object, wins where the partition limits regional naming (MNIST: 72.2 versus 44.4%). Payload and partition fit, not sensitivity, choose the interface.

c) Finding 3: publication evidence is a separate resource: On held-out clients charged to one total budget, a mean-gain certificate is cheap, accepting 86.8% of cases at $\varepsilon = 1$, but 12.7% of the classifiers it publishes recall some test class below 10%. Class evidence costs more, at a price set by the protocol. A naming share selected on independent development cohorts, frozen, and confirmed on new cohorts and the test split raises all-class acceptance from 5.4% to 78.2% at $\varepsilon = 1$ ($+22.8$ test accuracy points), makes recall $\geq .5$ certifiable at $\varepsilon = 3$ (76.2%), leaves mean-gain release unchanged and records no false accept. On natural FEMNIST writers, where no class target is reachable, a calculator prediction frozen before confirmation lands within 0.7 points in every cell, and 100.0% of the classifiers the mean-gain rule publishes recall some class below 10%.

d) Explaining the losses: Stored noise-free votes split the gap below the partition ceiling into signed differences: teacher content dominates on Usenet, release noise on CIFAR-1k and FEMNIST at $\varepsilon = .1$, selection on MNIST and CIFAR-10k. Private verification of a noise-gated rule screens cases, not tasks.

C. Contributions

- A resource-matched evaluation protocol with annotation access, privacy budget, payload and publication target as separate resources (Section II), and a map of which information supports deployment decisions (Section VII).
- Three findings on what to buy, release and verify, with a weaker-encoder ablation, published per-example and head baselines on the same protected sum, a cost ladder, tighter-band and source-disjoint controls, a budget split selected on independent cohorts and confirmed on CIFAR, and a Gaussian-approximation acceptance calculator tested on natural FEMNIST writers (Sections VI-B–VI-D).
- Signed differences below a decoder-matched partition ceiling that locate the loss on four tasks, and a pre-registered study of gated selection plus verification (Section VI-E).
- A public artifact that regenerates every number from immutable records and replays protected outputs of retrained cohorts.

II. Problem Formulation

A. Data roles

A coordinator holds a public encoder g, an unlabelled *proxy pool* P on which regions are formed, a *development pool* whose labels are revealed only under a declared acquisition mode, and a *test pool* used only offline. The class count and taxonomy are public; class frequencies and external validation labels are not free.

Private data are held by n clients. Client i holds a labelled dataset D_i and may train a local model f_i; the coordinator observes clients only through a protected sum over all participants, and the unit of protection is a complete client dataset. For private verification, each client also holds held-out records used only for bounded validation reports.

B. Resources

Each configuration carries a resource vector. *Annotation*: m distinct development labels and an acquisition mode, either *random* (uniform draws), *balanced* (m/C per class) or *feature-directed* (examples nearest each public region, requested before any label is revealed); every fitting, selection and public publication decision uses labels within m. *Privacy*: one total (ε, δ) budget under replace-one whole-client adjacency, composed over all releases of a deployment. *Payload*: integer coordinates per client and release, before secret sharing. *Public computation*: decoding, assignment, selection and an optional public student. Configurations differing in more than one resource are reported as separate conditions.

C. Release objects and endpoints

A release object is the protected vector a client contributes: a clipped parameter update, per-example proxy votes, or one bounded opinion per public region. The published endpoint is a *public classifier*; client models are never published.

D. Marginal value of private supervision

Let $\mathcal{H}_{\text{pub}}$ be a fixed family of public learners, each of which maps a label set L_m and the public features to a classifier. The encoder is part of the resource vector: a zero-shot head on another encoder, or one supervised by class names, changes the representation, not m, and is compared as a separate condition. Let $U(h)$ denote test accuracy (balanced accuracy for Usenet).

Definition 1 (Resource-matched marginal value)**.** For acquisition mode a, budget m and total privacy budget ε,

$$V_a(m,\varepsilon) = \mathbb{E}\big[U(h_{\text{joint}})\big] - \max_{h\in\mathcal{H}_{\text{pub}}} \mathbb{E}\big[U(h)\big], \tag{1}$$

where h_{joint} combines the same label set with one protected release under budget ε, and expectations run over label draws from mode a and over cohort/noise seeds.

No within-budget selection rule is guaranteed to reach the maximum over learner *means*, so Eq. (1) is conservative. It depends on a as well as m, and since a deployment sees one label draw and one cohort, we also report the per-draw gain distribution.

E. Publication targets

A positive V_a concerns average utility; a publishing coordinator may require more. We consider two targets for a candidate f against a fixed public comparator b. The *gain target* requires the finite held-out cohort mean $G = n^{-1}\sum_i [u_i(f) - u_i(b)]$ to be positive, where u_i is held-out accuracy at client i. The *classwise target* additionally requires every client-weighted class recall $\sum_i \mathrm{TP}_{i,c}/H_i \,/\, \sum_i N_{i,c}/H_i$ to be positive (H_i held-out records at client i, $N_{i,c}$ of class c); it is the pooled recall when all H_i are equal, as in the CIFAR studies. Evidence can come from public labels or from protected private statistics charged to the same total budget.

a) Research questions: They follow the platform's three decisions. **Q1** (buy): how does $V_a(m,\varepsilon)$ depend on acquisition mode and budget, and how is the per-draw gain distributed? **Q2** (release): does report sensitivity rank regional, per-example and head releases? **Q3** (verify): what does each publication target cost in total budget, and can the budget split be chosen before deployment? We then ask where, below the partition ceiling, regional value is lost. Sections VI-B, VI-C, VI-D and VI-E answer them in turn.

III. Release Interfaces and Public Readouts

A public partition defines a compact naming task. We compare protected regional votes, point votes and clipped heads, then hold their public readout and annotation resources explicit. The regional mechanism is called SDD in the reproducibility scripts; its direct NN readout is a naming classifier. Figure 2 shows one deployment: the platform fits the public partition, releases one protected regional matrix, decodes it jointly with the counted labels, and, when verification is required, releases a protected paired gain and a class histogram computed on client holdouts before publishing the candidate or the public comparator; all releases share one total budget.

A. A public partition for knowledge reuse

A fixed public extractor g and public K-means form nonempty regions $P_1,\dots,P_K$, shared across rounds independently of private data. Singleton regions recover per-example transfer; random partitions share the sensitivity, so matched controls test geometry.

B. One bounded opinion per region

For client model f_i, the regional posterior is

$$\bar{R}_i[k,\cdot] = \frac{1}{|P_k|}\sum_{x\in P_k} f_i(x) \in \Delta^{C-1}. \tag{2}$$

M votes for the largest component of $\bar{R}_i[k,\cdot]$. R votes on $f_i(x_k^*)$, where x_k^* is the public point nearest the centroid: public representative queries followed by regional label propagation. Both emit K one-hot rows over identical regions, matching sensitivity, payload and partition quality.

C. Noisy consensus and local learning

Each client adds independent integer discrete-Gaussian shares to its quantised report. The aggregate-only layer exposes $S = \sum_i (sR_i + G_i)$, and the server projects $S/(sn)$ row-wise onto the simplex to obtain A. For ordinary consensus, the target is

$$\hat{y}(x) = \arg\max_c A[k(x), c]. \tag{3}$$

Centered decoding uses a fixed class-prior correction: with public region weights $w_k = |P_k|$, subtract $\pi_c = \sum_k w_k A[k,c] / \sum_k w_k$ before taking the arg-max. It counters a common class preference under narrow local label coverage and is the fixed zero-label decoder. Raw and coverage decoding are separate post-processing rules of the same matrix; counted public labels can choose their use in joint naming.

a) Coverage-constrained decoding: At taxonomy resolution $K = C$, a prespecified alternative is a one-to-one assignment using the released matrix:

$$\hat{\pi} = \arg\max_{\pi\in\mathfrak{S}_C} \sum_k |P_k| A[k,\pi(k)], \qquad \hat{y}(x) = \hat{\pi}(k(x)). \tag{4}$$

Hungarian assignment solves it; it guarantees class coverage, while correctness depends on whether the public regions support a one-to-one class allocation. It needs no further private query or budget. Soft-target controls reuse this protected matrix and are separately indexed.

The primary $T = 1$ protocol trains each client privately before a single release. We compare 60 and 180 pre-release SGD steps without proxy targets. The naming-only contrasts allocate the full budget to this release; Section V-C allocates part to private verification. The $T = 3$ extension trains for 60 steps before each release, mixing equal private/proxy halves after the first. Releases 1/2 supervise later rounds; release 3 follows the final local fit. All three consume budget. The extension couples later teachers to earlier names. A private-only three-release control separates that feedback path from post-release averaging.

b) Two readouts from the same release: The public classifier predicts $\hat{y}(x) = \hat{y}_{\arg\min_k \|g(x)-\mu_k\|_2}$ using released regional labels and fixed public centroids, post-processing publishable under the transcript's whole-client DP; client models are not published. An optional public softmax student trains for 180 server steps on the protected labels only, with its public compute reported separately.

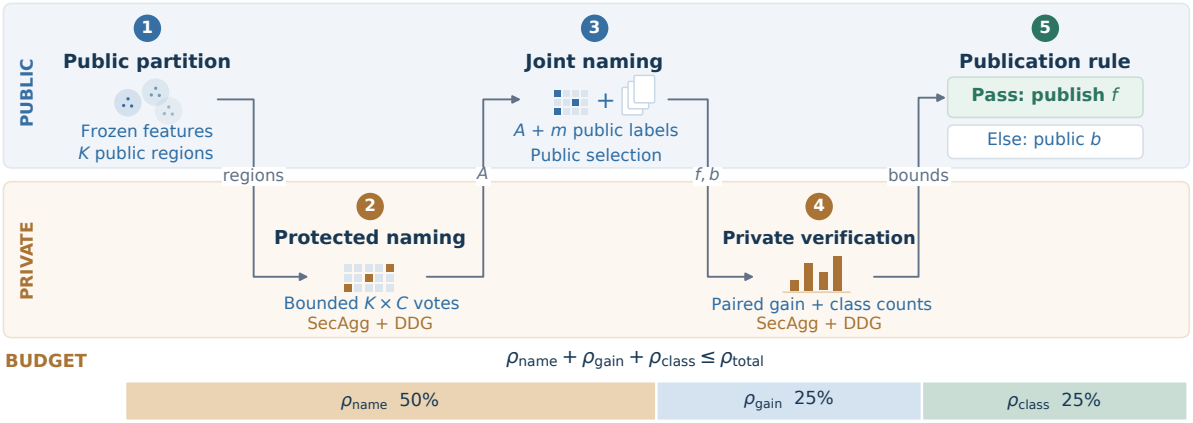


Fig. 2. One deployment across public and private operations. Protected naming (2) and verification (4) consume the same whole-client privacy budget; all operations on released statistics are public post-processing. Release objects (regional votes, clipped head, pooled per-example votes) differ only in step 2. The reference $50/25/25$ split is shown; other allocations change the shares, not the composition rule.

c) Payload and computation: At $|P| = 2000$, $C = 10$, a $K = 10/20$ release uses $100/200$ integer coordinates, versus 20000 per-example and 1280 head coordinates, before secret-sharing overhead. All server work after the release is public post-processing. Row-wise simplex projection of the $K \times C$ matrix costs $O(KC \log C)$ by sorting; centered decoding costs $O(KC)$; the coverage-constrained and joint decoders solve one $C \times C$ assignment in $O(C^3)$ time by the Hungarian method. Exact LOO for joint assignment needs at most $C+1$ assignment solves per weight (Proposition 1), independent of m. Server cost thus scales with K and C, not with private data size.

Taxonomy resolution $K = C$, M and centered decoding specify the main condition without public task labels; larger K and full-label configurations are separate conditions.

IV. Privacy Model and Analysis

A. Protected aggregate and whole-run client privacy

A fixed cohort of n clients participates in each of T rounds. Replace-one adjacency changes the entire dataset of one honest client. Public features, partitions and configurations are independent of private datasets. The server's entire view of client data is a sequence of protected sums and broadcasts; local models, reports and honest noise shares stay hidden.

a) Aggregation assumptions: A1: secure additive aggregation reveals the sum and metadata that is public or independent of protected data. A2: at most $t < n$ contributors disclose reports and noise to the server, leaving $h = n - t$ honest independent shares. A3: aggregation parties preserve that view against the stated coalition. A4: honest contributors use bounded reports and fresh independent noise; participation is fixed ($q = 1$), with recalibration or abort if the required honest cohort is unavailable. Secret-sharing or split-and-mix protocols can instantiate the interface [1]–[3]; experiments implement protected-sum arithmetic.

TABLE I
Release Interfaces under Replace-One Client Adjacency

released object	coordinates	L_2 sensitivity
clipped parameter update	d_{model}	$2B$
joint per-example votes	$\lvert P\rvert C$	$\sqrt{2\lvert P\rvert}$
public-region votes	KC	$\sqrt{2K}$

B. Sensitivity of the release objects

Table I compares clipping (norm B) and release resolution before quantisation and transport sharing; the compact head shares 128-dimensional features, while regional naming keeps each client's parameters local.

Lemma 1 (Report sensitivity). *For $R, R' \in (\Delta^{C-1})^K$, $\|R - R'\|_2 \le \sqrt{2K}$, with equality for opposing one-hot rows. This holds for every public partition.*

Lemma 2 (Integer sensitivity). *One-hot reports have $\Delta_q = s\sqrt{2K}$ exactly. Rounded posteriors have $\Delta_q \le s\sqrt{2K} + \sqrt{KC}$. A parameter update clipped at B and guarded after rounding in the integer ball $L_B = \lceil sB + \sqrt{d}/2 \rceil$ has $\Delta_q \le 2L_B$.*

Both follow because simplex rows differ in squared norm by at most two and rounding adds at most $\sqrt{KC}$; a guarded head lies in an integer ball of diameter $2L_B$.

C. Release mechanism and accounting

a) Release: Each client submits shares of $sR_i + G_i$ modulo 2^{64}, with $s = 2^{16}$ and independent coordinates $G_i[j] \sim \mathcal{N}_{\mathbb{Z}}(0, b^2)$. Noise is drawn by exact rejection sampling [4], so the analysis applies to the sampled distribution, not a floating-point approximation.

Theorem 1 (One protected aggregate). *Under A1–A4, the aggregate in dimension d is ρ_r-zCDP, where*

$$\rho_r = \frac{\Delta_q^2}{2hb^2} + d\,\tau_h(b), \quad \tau_h(b) \le 10(h-1)e^{-\pi^2 b^2}. \tag{5}$$

For $h = 1$, $\tau_h = 0$. The bound needs $b \geq 1/2$; every recorded release has $b \geq 1241$.

After disclosed coalition shares are subtracted, h honest shares remain, and Kairouz et al. [5, Prop. 14, full version] give ρ_r with $\tau_h = 10 \sum_{k<h} e^{-2\pi^2 b^2 k/(k+1)}$, which $k/(k+1) \geq \frac{1}{2}$ bounds as in Eq. (5); modular addition, projection, matching and soft-target construction are post-processing.

Theorem 2 (Adaptive whole run)**.** *Conditionally calibrated releases compose to $\rho_{\text{total}} \leq \sum_{r=1}^{T} \rho_r$. The complete server transcript is (ε, δ)-DP whenever*

$$\rho_{\text{total}} + 2\sqrt{\rho_{\text{total}} \log(1/\delta)} \leq \varepsilon. \tag{6}$$

Conditioning on data-independent random tapes and each preceding public transcript fixes the other clients' responses while the replaced report stays bounded, so adaptive zCDP composition covers every broadcast [6].

Test metrics and clean-count diagnostics are offline measurements, outside the deployment transcript claimed to satisfy this guarantee. Verification releases in Section V-C are further rounds of the same transcript and compose under Theorem 2.

D. Integer calibration and numerical certificates

Writing $L = \log(1/\delta)$ and $\rho_\varepsilon = (\sqrt{L+\varepsilon} - \sqrt{L})^2$, the total target is ρ_ε; for each per-release share ρ_j we use

$$b_j = \left\lfloor \Delta_{q,j} / \sqrt{2h\rho_j} \right\rfloor + 1, \tag{7}$$

and verify $\Delta_{q,j}^2/(2hb_j^2) + d_j\tau_h \leq \rho_j$. In every recorded release $\tau_h < 10^{-6.6\times 10^6}$, below the $10^{-30}/4$ the accountant charges, so all 47355 recorded certificates hold with this constant (artifact check). All budgets are recomposed with 90-digit arithmetic, and a union bound certifies modulo-2^{64} wrap probability below 10^{-30} per release.

Every release keeps a certificate with actual n, honest h, integer sensitivity, noise scale and composed budget. For $t = 0$, mean-noise standard deviation is bounded by

$$v = \frac{b\sqrt{n}}{sn} \approx \frac{\sqrt{TK/\rho_\varepsilon}}{n}. \tag{8}$$

Noise grows with $\sqrt{TK}$ and shrinks with n, so resolution, release count and cohort size can be traded before any private data are touched.

V. Label Budgets, Joint Naming and Verification

A. Annotation resources and joint naming

L_m holds the m labelled development examples of Section II; test queries are inductive and outside training graphs.

Fourteen public learners include regional majority, three assignment variants (differing in how empty regions are treated), class centroids, five ridge heads and four graph propagation configurations [7] with two strengths, with/without class-mass normalization. Ridge and graph learners use exact LOO predictions; a fixed public-first order resolves ties. Web selection uses observed-class balanced accuracy.

With observed regional label counts $L[k, c]$, projected private matrix A and public region sizes w_k, joint naming solves

$$\hat{\pi}_\lambda = \arg\max_{\pi \in \mathfrak{S}_C} \sum_k \left\{ L[k, \pi(k)] + \lambda \frac{w_k}{\bar{w}} A[k, \pi(k)] \right\}. \tag{9}$$

Here $K = C$, $\bar{w} = \sum_k w_k/K$ and $\lambda \in \{.01, .03, .1, .3, 1, 3, 10, 30\}$. LOO selects among the fourteen public learners, joint assignments and fixed private centered/coverage rules. All candidates reuse one protected matrix, so selection adds no private query. Exact LOO is efficient for the assignment candidates.

Proposition 1 (LOO on matched edges)**.** *For $S = L + \lambda\,\mathrm{diag}(w/\bar{w})A$, let π maximize assignment weight. Removing an example on an unmatched edge (k, c), $c \neq \pi(k)$, cannot make that edge correct in the reduced problem. Thus LOO correctness for all m examples needs at most $C+1$ assignment solves per weight.*

Every assignment using an unmatched removed edge loses one unit while the optimum loses none; examples on one matched edge share one refit.

The same joint decoder receives a protected head's mean proxy-region softmax or point scores averaged inside each region before projection. Native head prediction is an additional fixed head-arm candidate.

B. Public evidence and publication targets

The public augmentation rule requires a positive paired LOO lower gain, all-class proxy predictions and a positive Wilson-form recall bound per class (heuristics, since prediction and selection are dependent), else it falls back to the public model. A missing labelled class forces rejection, so verification labels also strengthen the public competitor; private verification changes the source of evidence and charges its privacy.

C. Private verification within one total budget

After protected naming fixes candidate f and public comparator b, client i computes held-out gain $d_i = u_i(f) - u_i(b) \in [-1, 1]$ and a normalized $C \times 4$ histogram of true class and joint correctness of f and b; both use distributed discrete-Gaussian shares.

The demonstration fixes ρ fractions $1/2, 1/4, 1/4$ for naming, gain and class statistics. Conditional composition in Theorem 2 charges all three releases to the same total $(\varepsilon, 10^{-5})$ budget. With released mean $\tilde{r}$ and gain-channel mean-noise proxy v_g, set

$$\underline{G} = 2\tilde{r} - 1 - 2v_g\sqrt{2\log(1/\alpha_g)} - s^{-1}. \tag{10}$$

Discrete-Gaussian sub-Gaussian tails [4] imply $\Pr(\underline{G} > G) \leq \alpha_g$, where $G = n^{-1}\sum_i d_i$. Direct histogram numerator/denominator bands yield simultaneous class-recall lower bounds (Proposition 2 below). We fix $\alpha_g = \alpha_h = .025$: gain-only acceptance uses $\underline{G} > 0$; class-checked acceptance also requires every recall lower bound positive. A *false accept* is an accepted case whose clean held-out statistics miss the target

($G \leq 0$, or some class recall zero or below the threshold). Rejection returns the public comparator without certifying its class recall. Coverage concerns validation noise and the finite held-out cohort, not unseen clients; publishing or rejecting is post-processing.

a) Bounded validation reports: In the cost ladder each client has 120 held-out records. For paired gain, encode $q_i = \text{round}(s(1 + d_i)/2) \in [0, s]$. Its integer sensitivity is s, and converting the rounded mean to gain adds at most $1/s$ error. The class report $h_i \in \Delta^{4C-1}$ records true class and joint binary correctness (a, b) of candidate/comparator. Largest-remainder rounding gives $Q_i \geq 0$, $\sum Q_i = s$, with coordinate error below $1/s$. Its integer squared sensitivity is exactly $2s^2$. Both reports have fixed dimension (1 and $4C$), independent of the number of held-out records.

Proposition 2 (Conditional finite-heldout coverage)**.** *Assume the specified reports and independent noise shares. Fix the naming transcript, public labels and actual held-out cohort. Equation* (10) *has one-sided coverage at least* $1 - \alpha_g$*. Let* $z = \sqrt{2\log(4C/\alpha_h)}$ *and define*

$$a_N = \sqrt{2}v_h z + 2/s, \qquad a_D = 2v_h z + 4/s. \tag{11}$$

For released normalized histogram $\tilde{h}$*, take* $D_c^U = \min(1, [\sum_{a,b} \tilde{h}_{c,ab} + a_D]_+)$ *and* $N_c^L = [\tilde{h}_{c,10} + \tilde{h}_{c,11} - a_N]_+$*. The clipped ratio* N_c^L/D_c^U *is simultaneously a recall lower bound for all classes with probability at least* $1 - \alpha_h$*, taking zero if* $D_c^U = 0$*. Joint gain/class coverage is at least* $1 - \alpha_g - \alpha_h$*, apart from the stated wrap term.*

Proof sketch. Each exact share is sub-Gaussian with proxy b (Lemma 16 of Canonne et al. [4]), so the mean has proxy v; a correct-mass sum has proxy $\sqrt{2}v_h$ and a class mass $2v_h$. A union over the $2C$ fixed linear targets with rounding errors gives Eq. (11), and monotone division plus a union with the gain event completes the claim. □

This is noise coverage for a finite cohort, not a posterior probability given acceptance. Whole-client adjacency affects training and held-out records, so all releases compose sequentially.

b) Why targets differ in cost: The gain target resolves one scalar with multiplier $\sqrt{2\log(1/\alpha_g)}$; classwise targets resolve $2C$ simultaneous bands with the larger z, and each must separate from zero a class mass of order $1/C$. Section VI-D measures the resulting cost ladder.

Higher K can raise purity while increasing the noise in Eq. (8); natural writers test this.

VI. Experimental Evaluation

The evaluation follows the platform's three decisions (Q1–Q3), then explains where regional value is lost; every comparison states its resource vector.

A. Tasks, resources and replication

Table II lists the settings (features: encoder/proxy-fitted dimension; splits: proxy/development/test) and Table III the studies, their cohorts and what each froze before its outcomes. Images use frozen DINOv2 ViT-S/14 features [8] and Usenet public TF-IDF, each reduced by proxy-fitted PCA/SVD and row-normalized. Constructed image clients draw 600 images with replacement under Dirichlet $\alpha = .1$; the protected unit is a complete constructed reporting dataset, and source copies at other clients remain in the adjacent database. Usenet clients are From-header authors over six content families, with headers and quotes removed, disjoint authors per split, and balanced accuracy as metric. Official 62-class FEMNIST [9], [10] assigns distinct writers, with their original records, to private, public and test roles. All recalls are recorded. These task labels can be supplied by public annotators; no claim of irreplaceable private Web semantics follows.

TABLE II
Evaluated Settings

Task	C	Features	Private	n	ε	Splits
CIFAR	10	DINOv2/128	45000 img.	1k, 10k	.1	2000/5000/4033
MNIST	10	DINOv2/128	53000 img.	10k	.1	2000/5000/5000
Usenet	6	TF-IDF/128	4249 authors	3k	1	1500/1500/1737
FEMNIST	62	DINOv2/128	2600 writers	1k, 2k	.1, 1	2000/5000/4549
Verif.	10	DINOv2/128	480/120 rec.	1k	.1, 1, 3	as CIFAR

TABLE III
Study Designs

Study (§)	Cohorts × label draws	Frozen first	Reported
Label access, interfaces (B, C)	10 protocol + 10 confirmation cohorts × 100	design	Tab. IV, Figs. 3, 4
Weaker encoder (B)	CIFAR-1k confirmation cohorts	design	text
Cost ladder (D)	10 cohorts × 100 × 3 ε (3000)	split, targets	Tab. V
Protocol price (D)	ladder cohorts, draws 50–99	Opt. post hoc	Tab. VI
Source-disjoint (D)	10 new cohorts × 100	design	Tab. VI
Split, CIFAR (D)	10 + 10 cohorts, disjoint sources (54000)	design, split	Tab. VII
Split, FEMNIST (D)	10 + 10 cohorts, disjoint writers (27000)	split, planner	text
Gated + verified (E)	10 new cohorts per task (6000)	design	Tab. IX
Ceilings (E)	clean votes of label-access cohorts	offline	Tab. VIII, Fig. 6

a) Protocols and phases: The artifact lists all cohort phases and seeds. Softmax learners use batch 128, learning rate .1, momentum .9 and weight decay 5×10^{-4}. Naming-only T=1 releases once after 60/180 private steps; feedback T=3 uses 60 steps before each release. All set $\delta = 10^{-5}$ with equal actual/calibrated populations. Heads fix clipping at $B = 1$ without task labels.

For $m/C \in \{1, 3, 5, 10, 20, 50\}$, random acquisition has 100 nested label draws; CIFAR/MNIST balanced acquisition has 50 draws. Feature-directed annotation selects q nearest development points per public region before revealing labels, filling shortages by nearest-to-assigned-centroid points. Each set crosses ten cohort/noise seeds. A two-way bootstrap (2000

resamples) resamples label draws and cohorts; simultaneous public gain intervals cover all six budgets and fourteen public learners.

b) Implementation and reproducibility: Graphs use 20 neighbors and $\alpha = .8/.95$; ridge uses .001–10; k-means uses five starts and seed 1729. Research noise keys and clean diagnostics are outside the deployment transcript; one GPU computes the protected-sum arithmetic, not secure-aggregation performance. Cases sharing cohorts and label sets are not independent trials. The artifact maps every table and figure to its records; `verify` re-aggregates them and re-checks every budget certificate, and `smoke` retrains cohorts and reproduces their protected outputs and decisions.

B. What to buy: annotation access and the gain distribution

With balanced labels (50 draws; the artifact tabulates all fourteen public learners), three supplied labels per class name the fixed CIFAR partition at 91.59% in all 50 draws (graph propagation: 92.61%), and ten feature-directed annotations recover the same classifier without private reports. On MNIST's less class-aligned regions one label per class gives 38.28% majority naming versus 58.36% graph propagation.

Under passive random access, independent CIFAR-1k with ten labels gives 61.62% for the strongest mean public learner (raw-count matching) and 74.62% for joint naming (Figure 3). The 13.0-point mean gain has simultaneous interval [4.6,21.4]. The median label-mean gain is 13.6 points, so a few large draws do not drive it; its fifth percentile is -3.8, and 9% of label means and 16% of label/cohort cases are negative (Figure 4(b)). Label- and cohort-mean SDs are 10.11/7.95 points.

Every ten-label CIFAR draw omits a class (30/50 labels: $31/7\%$), and the public class-check rule rejects private augmentation at every tested budget, so 74.62% is an ungated utility (the artifact tabulates all fallbacks). Random-only access is an experimental contract, not a known constraint of CIFAR or Usenet. LOO selection contributes its own variation: the public portfolio gives 56.13% versus fixed count matching's 61.62%, and at CIFAR-10k joint selection gives 87.35% versus fixed zero-label private naming's 91.59% (Section VI-E).

a) A weaker encoder: A frozen design replaces DINOv2 by an ImageNet ResNet-18 and reruns the independent CIFAR-1k comparison with the same images, cohorts, label draws, learners and release. The partition ceilings fall to 61.9/62.2%. Ten feature-directed labels again reach the one-to-one ceiling without private reports (61.9%); with ten random labels, joint naming (38.6%) changes the strongest public learner (42.6%) by -4.0 points ([-5.4,-2.6]). The registered E1, a positive passive-access gain, fails; E2, directed labels beating random labels plus private reports, holds. Under both encoders acquisition decides whether private reports are needed; their passive-access value also needs regions that carry the classes (Section VI-E).

C. What to release: sensitivity versus accuracy

Per-example votes are $\sqrt{|P|/K}$ times more sensitive than regional reports (Section IV); at $|P| = 2000$ and $K = 10$ the factor is about 14, and they carry 20000 rather than 100 coordinates per client, against 1280 for a head. Table IV holds post-processing fixed: all three objects feed the same joint decoder and LOO selection with the same labels (I/P: independent/paired cohorts; Public: strongest of the fourteen public learners; Native Head uses no labels). Pooled point and regional joint naming differ by at most 2.3 points on all 8 settings (CIFAR-1k: 76.94 versus 74.62%), while the head, whose native predictions the partition does not bound, wins where the partition limits regional naming (MNIST-10k: 72.2 versus 44.4%; at CIFAR-10k native Head gives 94.0%). Published designs run through the same protected sum in the inherited three-release protocol (10^4 clients, $\varepsilon = .1$, 5 seeds, publicly configured settings): on CIFAR-10, DP-FedCT's per-example votes, our Point interface, reach 69.7% and PATE-DI's screened teacher votes (RDP conversion) 80.5%, against 85.0% for regional reports and 92.7% for the clipped DP-Head (MNIST: 46.2, 52.0, 55.0 and 67.3%).

Pooling explains why the sensitivity gap does not reach accuracy. Relative to regional noise, averaging independent point noise in region k has standard-deviation ratio approximately

$$\frac{v_{\text{point,pool},k}}{v_{\text{region}}} \simeq \sqrt{\frac{|P|}{K|P_k|}}. \tag{12}$$

CIFAR's region sizes give ratios .88–1.07: pooled point votes carry about the regional noise. At zero labels, CIFAR-1k regional-minus-pooled paired differences are $+6.39$ points ($[-10.48, 23.26]$) in protocol cohorts and -8.02 ($[-26.81, 10.77]$) in independent cohorts (Figure 4(a)); neither supports a stable regional advantage. The $\sqrt{|P|/K}$ ratio is a payload and interface property, not an accuracy guarantee.

With fixed 60-step teachers, T=1 raises CIFAR-1k from 26.86% to 49.56%; no-feedback T=3 controls show no universal feedback penalty. The pooling, round and no-feedback tables are in the artifact.

D. What evidence to buy: mean gain versus class evidence

The verification demonstration (Table II, Verif.) uses ten new CIFAR-1k cohorts, 100 fresh ten-label draws and an 80/20 source split between training and held-out records. The comparator is raw-count matching; candidates (eight joint weights, fixed private and public count rules) are chosen from protected naming and the ten labels. Each of the 3000 cases charges all three releases to one total budget; every utility is unconditional, a rejection scoring its public fallback.

a) A cost ladder of publication targets: Table V evaluates practical targets on these protected outputs (acceptance / unconditional test accuracy; each row is a separately declared rule with its own $\alpha = .025$ class family). Gain-only acceptance is 4.6% at $\varepsilon = .1$ and 86.8% at $\varepsilon = 1$, where direct class-sum bounds accept 6.2% (initial coordinate-wise bands: zero). At $\varepsilon = 3$, class-checked acceptance reaches 75.9% (86.63% test accuracy against 62.30% public), a budgeted route to classwise evidence, not a validation of the 74.62% classifier at unchanged privacy. A mean gain of at least .20 still accepts 58.2% of cases at $\varepsilon = 1$. All-class recall of

TABLE IV
EQUAL-LABEL COMPARISON OF RELEASE INTERFACES

Task, clients / ε	Group	Public	Region joint	Head joint	Point joint	Native Head
CIFAR 1k / 0.1	I	61.6	74.6	76.6	76.9	34.3
CIFAR 10k / 0.1	I	61.6	87.4	91.1	87.4	94.0
MNIST 10k / 0.1	I	46.2	44.4	72.2	44.7	77.0
Usenet 3k / 1	P	32.1	28.2	31.8	28.9	23.2
FEMNIST 1k / 0.1	P	37.3	36.0	36.0	36.0	12.1
FEMNIST 1k / 1	P	37.3	35.9	38.1	36.0	40.7
FEMNIST 2k / 0.1	P	37.3	36.0	35.9	36.0	26.3
FEMNIST 2k / 1	P	37.3	35.8	38.5	36.1	41.3

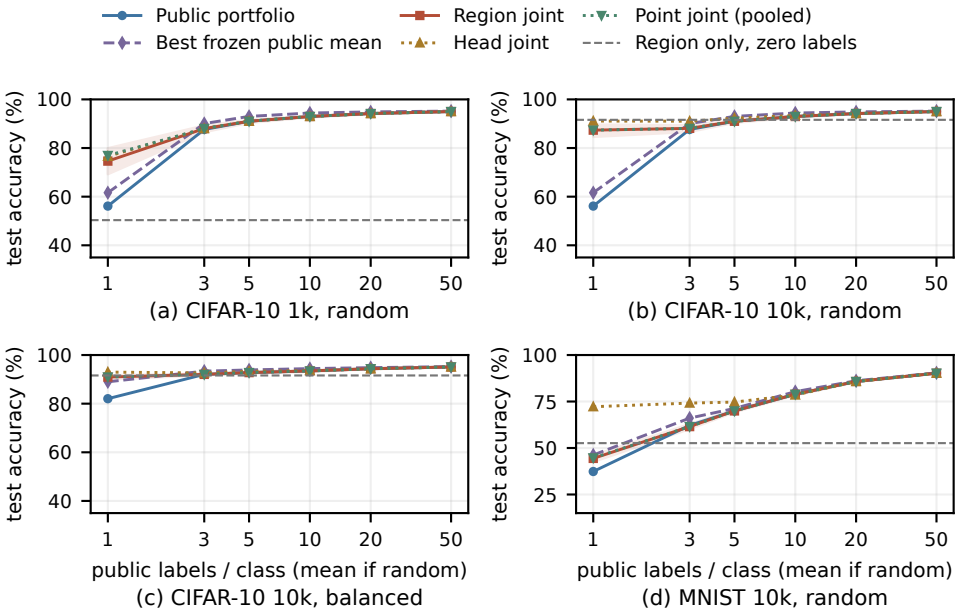


Fig. 3. Label access changes the comparison. Random panels use independent confirmation cohorts/labels; balanced uses protocol cohorts. All private arms release once at $\varepsilon = .1$ after 180 private steps. Public envelope is descriptive over fourteen frozen learners; the portfolio is a LOO rule. Private joint curves are ungated. All fitting/selection labels are inside m. Shading is crossed pointwise bootstrap; text gives simultaneous intervals.

at least .25 is accepted in 75.7% of cases at $\varepsilon = 3$, and .50 at no tested budget. Per-class non-degradation against the comparator, $\text{recall}_{f,c} - \text{recall}_{b,c} \geq -\eta$, depends on its tolerance: at $\varepsilon = 3$ it accepts 2.0% at $\eta = .25$ but 83.1% at $\eta = .5$; its band radius must fall below η times a class mass of order $1/C$. Costs are ordered only for stated thresholds. Gain-policy false accepts are 1/1000 at $\varepsilon = .1$, and no ladder rule records one at $\varepsilon \geq 1$.

b) What the cheap target publishes: A correct mean-gain certificate says nothing about the worst class (Figure 5(a)). At $\varepsilon = 1$ the mean-gain rule publishes 868 of 1000 cases; 6.6% of them recall some class below 1% and 12.7% below 10% (12.8% at $\varepsilon = 3$). Every classifier the all-recall rule publishes recalls each class above 50%; the ladder's price difference buys exactly this tail.

c) Protocol price versus target price: The ladder fixes one protocol: a $50/25/25$ split, both reports released and two-sided bands. Table VI varies each choice on draws 50–99. *Tight* applies one-sided lower bands to the same histograms (all-recall acceptance at $\varepsilon = 1$: 6.4% to 19.8%, nothing else changes); *Re-rel.* re-releases the stored clean statistics with fresh exact noise after reconstructing all 3000 histograms bit for bit; *Opt.* uses the naming share ϕ that maximized offline test accuracy on draws 0–49 (gain: ϕ to naming, the rest to the gain report; classes: the gain report takes a quarter of the rest). Both phases score the same test set and cohorts, so *Opt.*

TABLE V
VERIFICATION-COST LADDER OF PUBLICATION TARGETS

Publication target	$\varepsilon = .1$	$\varepsilon = 1$	$\varepsilon = 3$
gain > 0	4.6 / 63.7	86.8 / 88.9	87.0 / 88.9
gain $\geq .10$	2.2 / 63.0	82.0 / 88.2	85.0 / 88.7
gain $\geq .20$	0.7 / 62.6	58.2 / 83.4	64.6 / 85.1
all recalls > 0	0.0 / 62.3	6.2 / 64.3	75.9 / 86.6
all recalls $\geq .25$	0.0 / 62.3	0.0 / 62.3	75.7 / 86.6
all recalls $\geq .50$	0.0 / 62.3	0.0 / 62.3	0.0 / 62.3
non-degr. $\eta = .25$	0.0 / 62.3	0.0 / 62.3	2.0 / 63.2
non-degr. $\eta = .50$	0.0 / 62.3	0.0 / 62.3	83.1 / 87.8

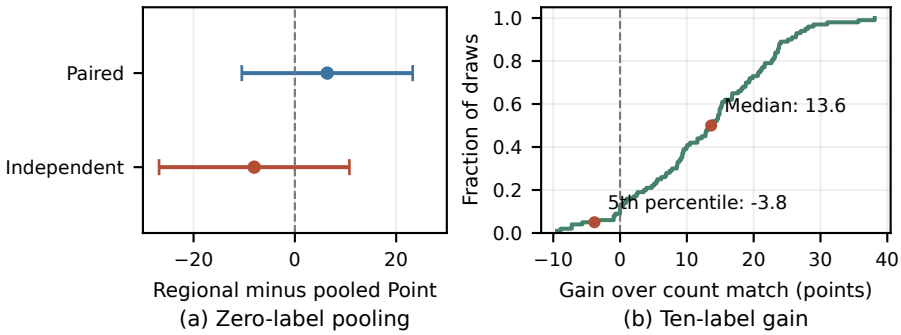


Fig. 4. CIFAR-1k, $\varepsilon = .1$. (a) Zero-label centered NN: regional/pooled Point means are $52.2/45.8\%$ in paired cohorts and $50.3/58.4\%$ in independent cohorts. Paired Student intervals use ten seeds per group. (b) Each label-mean gain averages ten independent-confirmation cohorts against fixed raw-count public matching. The 100 draws have mean 13.0, median 13.6 and fifth percentile -3.8 points. Crossed cases are not independent.

TABLE VI
VERIFICATION COST UNDER FOUR PROTOCOLS

Target	ε	Fixed	Tight	Re-rel.	Opt.	ϕ	Disjoint
gain > 0	.1	5.4/64.5	5.4/64.5	6.6/64.6	12.4/66.4	.4	5.0/64.2
	1	88.0/90.3	88.0/90.3	88.0/90.3	88.0/90.2	.1	88.0/90.2
	3	88.0/90.3	88.0/90.3	88.0/90.5	88.2/90.5	.5	88.4/90.6
all recalls > 0	.1	0.0/62.8	0.0/62.8	0.0/62.8	0.0/62.8	–	0.0/62.8
	1	6.4/64.9	19.8/69.0	7.8/65.2	81.8/88.5	.1	7.4/65.1
	3	82.8/89.0	82.8/89.0	83.6/89.5	83.6/89.3	.3	84.2/89.6
all recalls $\geq .50$	.1	0.0/62.8	0.0/62.8	0.0/62.8	0.0/62.8	–	0.0/62.8
	1	0.0/62.8	0.0/62.8	0.0/62.8	0.0/62.8	–	0.0/62.8
	3	0.0/62.8	0.0/62.8	0.0/62.8	70.2/85.0	.1	0.0/62.8

is a retrospective sensitivity analysis: the fixed split is far from best for class targets ($\phi = .1$: 7.8% to 81.8% at $\varepsilon = 1$), so Table V prices one protocol, not the targets.

d) Choosing the split before deployment: Can a platform find such a split without the test set? A third frozen design splits the 45000 private CIFAR images per class into two pools; ten *development* cohorts come from one and ten *confirmation* cohorts from the other, each with 1000 clients holding 16 training and 4 held-out records used once. On development cohorts and label draws 0–49, each target and budget gets the grid ϕ that maximizes the accuracy of the published classifier on the labelled public validation split; the selection reads only protected decisions and validation labels and was frozen before any confirmation case ran. Confirmation cohorts, label draws 50–99 and the test split then compare the frozen share with the fixed split at the same total budget (Table VII, 54000 cases; Δ with crossed-bootstrap 95% interval, FA false accepts, Best the unused confirmation-best share). The pre-registered primary test holds: at $\varepsilon = 1$, all-recall acceptance rises from 5.4% to 78.2% ($+72.8$ points, $[+62.0, +83.0]$) and test accuracy from 64.5 to 87.2% ($+22.8$, $[+18.9, +26.7]$). Recall $\geq .5$ becomes certifiable at $\varepsilon = 3$ (76.2% acceptance, 87.2% accuracy), mean-gain cells move by at most 0.1 points, and no frozen cell records a false accept. Development and confirmation curves agree (Figure 5(b,c)); the frozen share loses at most 1.5 points to the confirmation-best one. The registered prediction that class shares fall below .5 at $\varepsilon \leq 1$ failed only where no share certifies anything and ties default to .5. One development cohort with ten draws already selects well (86.2% confirmation accuracy at $\varepsilon = 1$, 84.2% for recall $\geq .5$ at $\varepsilon = 3$, mean of 50 such pilots). The pilot is a separate population with its own ledger: 100 development labels (ten draws of ten), the 3967 public validation labels that score each published classifier, and 267 protected releases over 17 grid configurations. Answered by one population they compose to 97.1 times a deployment's zCDP budget (total $\varepsilon \approx 11.7$ for deployments at $\varepsilon = 1$). The deployment's public learners do not see the pilot's labels; a platform allowed to train on its validation labels should compare against learners that do.

e) Natural writers: an external test of the planner: FEMNIST runs the same protocol (Table VII, lower block): 1000 writers per cohort from two disjoint pools, a seeded 80/20 split of each writer's records (H_i varies, median 32), 62 random labels, 27000 cases. Reports, certificates, clean statistics, planner inputs and false-accept judgments all use the client-weighted statistics of Section II. Planner predictions frozen before confirmation lie within 0.7 points of every cell, with no false accept; the cells sit near 0 or 100%, so this checks calibration at the extremes. The study prices verification, not private value: the mean-gain rule accepts 99.0% at $\varepsilon = 1$ against the 11.6% count-matching comparator, yet its 35.1% stays below the strongest public learner on the same labels (37.2%) and 100.0% of what it publishes recalls some class below 10%. Candidates miss whole classes, so no class target is certifiable and the split cannot help, as the planner predicted.

f) Source-disjoint clients: Constructed CIFAR clients in the ladder reuse source images. A design frozen before implementation rebuilds the ladder on ten new cohorts in which each of 40000 private images belongs to one client, 32 for training and 8 held out, with the same candidates, labels, split and bands (Table VI, Disjoint). The mean top-class share is 0.672 (last 100 clients 0.667; 0.664 with reuse). The gain policy beats the fallback by 26.6 points at $\varepsilon = 1$ ([23.7,29.5]; 26.6 with reuse) and every rung stays within 1.4 points; the predicted shrinkage did not occur. Its two false accepts at $\varepsilon = 3$ have true gain zero, within α. Disjointness also cuts records per

TABLE VII
BUDGET SPLIT SELECTED ON DEVELOPMENT COHORTS AND CONFIRMED ON NEW COHORTS

Task	Target	ε	ϕ	Fixed	Frozen	Planner	Δ accuracy	FA	Best
CIFAR	gain > 0	.1	.3	3.8/63.8	4.0/63.8	–	+0.0 [−1.0, +1.0]	0	.5
		1	.5	88.0/90.2	87.6/90.0	–	−0.1 [−0.7, +0.2]	0	.6
		3	.2	88.0/90.6	88.0/90.6	–	+0.0 [−0.2, +0.2]	0	.4
	all recalls > 0	.1	.5	0.0/62.8	0.0/62.8	–	+0.0 [+0.0, +0.0]	0	–
		1	.2	5.4/64.5	78.2/87.2	–	+22.8 [+18.9, +26.7]	0	.3
		3	.2	83.8/89.6	83.6/89.5	–	−0.1 [−0.8, +0.3]	0	.6
	all recalls $\geq .50$	.1	.5	0.0/62.8	0.0/62.8	–	+0.0 [+0.0, +0.0]	0	–
		1	.5	0.0/62.8	0.0/62.8	–	+0.0 [+0.0, +0.0]	0	–
		3	.1	0.0/62.8	76.2/87.2	–	+24.4 [+19.9, +28.6]	0	.1
FEMNIST	gain > 0	1	.9	99.0/35.1	98.2/35.2	98.7/98.1	+0.0 [−0.7, +0.6]	0	.7
		3	.2	100.0/35.5	100.0/35.5	99.3/99.7	+0.0 [−0.6, +0.5]	0	.3
		10	.1	100.0/35.1	100.0/35.1	100.0/100.0	+0.0 [−0.2, +0.3]	0	.3
	all recalls > 0	1–10	.5	0.0/11.6	0.0/11.6	0.0/0.0	+0.0 [+0.0, +0.0]	0	–
	all recalls $\geq .50$	1–10	.5	0.0/11.6	0.0/11.6	0.0/0.0	+0.0 [+0.0, +0.0]	0	–

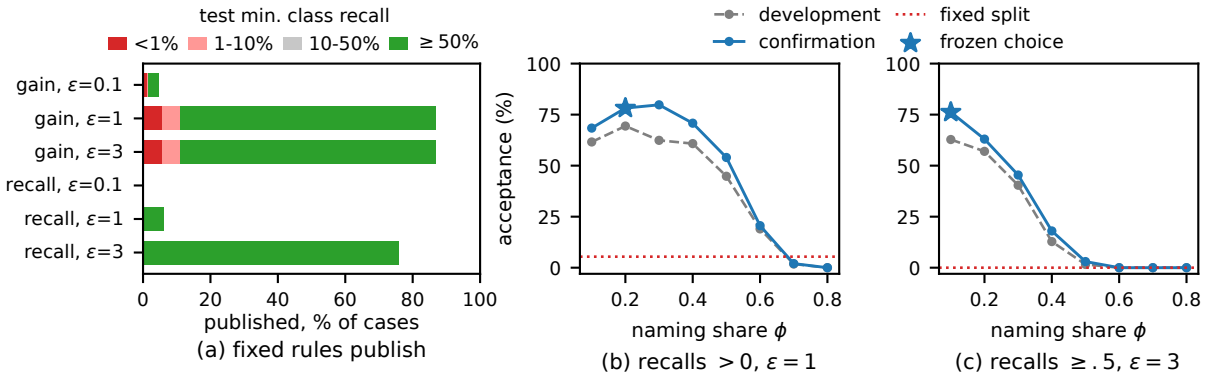


Fig. 5. (a) The 1000 cases per budget of Table V, by the test minimum class recall of the classifier each fixed rule publishes (offline diagnostic). (b, c) Acceptance over the naming share ϕ on the development cohorts (dashed, used for selection) and the confirmation cohorts (solid); the star is the frozen choice and the dotted line the fixed $50/25/25$ split on the confirmation cohorts.

client, so this does not show that reuse is irrelevant.

g) A planning map beyond the tested setting: Per case, the calculator takes the clean gain g, class masses m_c, recalls r_c and differences d_c. Treating each report as Gaussian with its calibrated std ($2v_g$ for the gain, v_h per coordinate) and keeping rounding terms, it returns $\Phi((g - a_g - \gamma)/2v_g) \prod_c \Phi(\mu_c/\sigma_c)$, with a_g the gain band radius. A class rule accepts when a linear statistic of class c clears zero: for recall $\geq r_0$, $N_c^L - r_0 D_c^U \geq 0$, so $\mu_c = r_c m_c - a_N - r_0(m_c + a_D)$ and $\sigma_c = v_h\sqrt{2(1-r_0)^2 + 2r_0^2}$ (Eq. (11)); for non-degradation, $\mu_c = (d_c + \eta)m_c$ minus its radius and $\sigma_c = v_h\sqrt{2 + 4\eta^2}$. The linear statistics ignore the clipping of the bands. Capping D_c^U at 1 can only raise acceptance and binds only when the released class mass exceeds $1 - a_D$ (probability $\Phi((m_c + a_D - 1)/2v_h)$); clipping at zero changes a decision only when that mass falls below $-a_D$. Outside these events the linear form is exact. Classes and reports multiply because their noise is independent. The result is an approximate planning quantity, not a coverage guarantee. It reproduces all 24 rates of Table V within 1.1 points; frozen before the new-cohort study of Section VI-E, it also matches its 12 gain and class rates (other n, C and natural Usenet clients, class masses 0.055–0.387) within 0.5 points. This validates the noise calibration, not effect sizes. At the median CIFAR-1k effect sizes, the artifact's planning map gives the smallest total ε accepting each target with probability one half; as $v \propto 1/(n\sqrt{\rho})$, the required ρ scales as n^{-2} and ε roughly as $1/n$. It is conditional on effect sizes: at $n = 10$k it prices all recalls > 0 at 0.13, yet the new MNIST candidates, with a nearly empty worst class, pass in 0.0% of cases at $\varepsilon = 1$.

E. Explaining the losses: partition ceilings

A regional readout assigns one class per public region, so its accuracy is bounded by the *partition ceiling*, the best such assignment within the decoder's family: $\text{Ceil}_{1:1}$ for assignment decoders, $\text{Ceil}_{M:1} \geq \text{Ceil}_{1:1}$ for centered decoding. With $P^\star$ the strongest public learner, the regional part of the marginal value in Eq. (1) for decoder family D satisfies

$$V_D^{\text{reg}} \leq \text{Ceil}_D - P^\star. \tag{13}$$

The stored clean vote matrices, never released, make the gap below the ceiling measurable. Decoding them with the centered decoder of the protected release gives *noise-free*

TABLE VIII
WHERE REGIONAL VALUE IS LOST

Task/ε	Ceil.	Pub.	Room	Clean	Named	Joint	Δ_T	Δ_N	Δ_S	N↑	v	Gated	Head
CIFAR 1k/0.1	91.6/91.6	61.6	+30.0	91.6	50.3	74.6	+0.0	+41.2	-24.3	0/10	0.215	74.6	76.6
CIFAR 10k/0.1	91.6/91.6	61.6	+30.0	91.6	91.6	87.4	+0.0	+0.0	+4.2	0/10	0.022	91.6	91.1
MNIST 10k/0.1	52.6/54.1	46.2	+7.9	53.9	52.6	44.4	+0.2	+1.3	+8.2	2/10	0.022	52.6	72.2
Usenet 3k/1	42.4/42.5	32.1	+10.3	32.3	20.7	28.2	+10.2	+11.6	-7.5	0/10	0.006	21.0	31.8
FEMNIST 1k/0.1	24.6/42.9	37.3	+5.5	35.3	7.1	36.0	+7.5	+28.3	-28.9	0/10	0.535	36.0	36.0
FEMNIST 1k/1	24.6/42.9	37.3	+5.5	35.3	35.5	35.9	+7.5	-0.2	-0.4	5/10	0.055	35.9	38.1
FEMNIST 2k/0.1	24.6/42.9	37.3	+5.5	35.3	19.8	36.0	+7.5	+15.5	-16.2	0/10	0.268	36.0	35.9
FEMNIST 2k/1	24.6/42.9	37.3	+5.5	35.3	35.6	35.8	+7.5	-0.2	-0.2	6/10	0.027	35.6	38.5

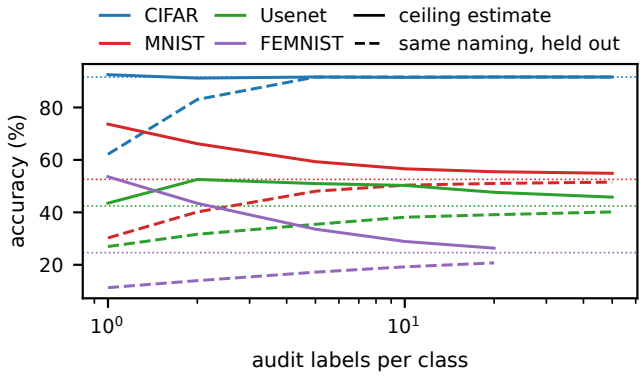


Fig. 6. Auditing the ceiling. A random labelled audit of the development split: its in-sample one-to-one ceiling estimate (solid) and the test accuracy of the naming it induces (dashed); dotted: true ceilings; 200 repetitions; offline.

naming N_0; with protected naming N_ε and joint selection J, on one partition, decoder, data and seed set,

$$\text{Ceil}_{M:1} - J = \underbrace{\text{Ceil}_{M:1} - N_0}_{\Delta_T} + \underbrace{N_0 - N_\varepsilon}_{\Delta_N} + \underbrace{N_\varepsilon - J}_{\Delta_S} \quad (14)$$

The identity telescopes, so its terms are signed controlled differences, not causal attributions: noise improves naming on 2/10 MNIST seeds and 6/10 FEMNIST-2k/1 seeds, and Δ_S is negative wherever fusion with the labels corrects noisy names (-24.3 points on CIFAR-1k). We read Δ_T as teacher content, Δ_N as release noise and Δ_S as selection only where a term is large and stable across seeds (Table VIII; Room is $\text{Ceil}_{M:1} - P^\star$, N↑ counts seeds where noise improves naming, v is the naming-noise std).

Teacher content bounds the natural-client tasks. Usenet has 10.3 points of room, but noise-free naming reaches only the public level, 32.3% ($\Delta_T = 10.2$): votes barely differ between regions, so even small noise ($v = 0.006$) flips most regions and costs another 11.6 points. On FEMNIST no assignment decoder can add value (one-to-one ceiling 24.6%, below public); the many-to-one ceiling leaves 5.5 points, and noise-free naming reaches 35.3% ($\Delta_T = 7.5$), which protected naming matches at $\varepsilon = 1$. *Release noise* adds 15.5–28.3 points on FEMNIST at $\varepsilon = .1$ and 41.2 points on CIFAR-1k ($v = 0.215$), where clean naming is at the ceiling and joint fusion with ten labels recovers 74.62%. *Selection* is the dominant term where protected naming already realizes the room. On CIFAR-10k ($v = 0.022$) fixed protected naming reaches the ceiling, 91.6%, and LOO loses 4.2 points. On MNIST the many-to-one ceiling is 54.1%; $\Delta_T = +0.2$, $\Delta_N = +1.3$ and $\Delta_S = +8.2$, turning a 7.9-point room into a negative joint gain, while the head, which the partition does not bound, reaches 72.2% (Section VI-C).

a) An exploratory selection rule: Random $m = C$ draws almost always miss a class, and LOO then compares candidates on classes it cannot see. After earlier rounds, we registered a rule family: LOO; keep protected naming whenever labels miss a class; or do so only if also the public noise scale $v(n, \varepsilon, K) \le v_0 \in \{.01, .03, .1\}$. One rule is selected on draws 0–49 and evaluated on draws 50–99 of the same cohorts, an exploratory estimate. The coverage-only rule fails where noise is large (-11.3 points on CIFAR-1k). The selected rule, $v_0 = 0.03$, raises the held-out gain over the strongest public learner from -2.2 to $+7.3$ points on MNIST and from $+24.2$ to $+27.7$ on CIFAR-10k, but lowers Usenet from -4.0 to -11.3: its noise is the smallest, and the rule cannot see its poor teacher. No tested rule dominates; choosing privately would be private selection [11].

b) Private verification screens the gated candidate: A frozen design runs on new cohorts (10 per task, 6000 cases): the gated rule names a candidate, the LOO portfolio is the comparator, and disjoint holdout clients release gain and class histogram under one $50/25/25$ budget. At $\varepsilon = 1$ the gate is open and its sign replicates ($+10.6$ points over LOO on MNIST, -4.4 on Usenet; Table IX, whose acceptance and false accepts (FA) count cases with a private candidate, Priv.). Verification accepts 99.6% of private candidates on MNIST and 98.0% on CIFAR-10k with no false accept in any case, so the verified rule keeps the MNIST gain ($+10.6$ $[+8.0, +13.6]$) and returns Usenet's mean balanced accuracy to the LOO level ($+0.5$ $[-0.7, +1.7]$), without a per-case guarantee. The pre-registered task-level criterion, at most 10% acceptance on Usenet, was not met: 20.6% of its 495 private candidates were accepted. Verification separates cases, not tasks. Post hoc, the 102 accepted Usenet cases gain $+15.8$ test accuracy points and the rejected ones -15.1 (draws where the six-label portfolio is weak), but the target is client-mean accuracy and accepted cases gain only $+2.1$ balanced points (62.7% positive). A balanced-accuracy rule on the same released histogram, added

TABLE IX
GATED RULE PLUS PRIVATE VERIFICATION ON NEW COHORTS

		Accept						Verified	
Task	Priv.	acc	bal	FA	Pub.	LOO	Gated	acc	bal
MNIST 10k	500	99.6	77.4	0	37.3	43.0	53.6	53.6	52.3
CIFAR 10k	500	98.0	98.0	0	55.9	88.0	91.6	91.6	91.6
Usenet 3k	495	20.6	0.0	0	28.2	28.2	23.8	28.7	28.2

TABLE X
WHAT THE PLATFORM CAN USE, AND WHEN

Information	Available	Use
Acquisition contract, report dimension, calibrated noise scale v, a pilot population's protected decisions and validation labels	before deployment	choose comparable configurations and the budget split
Protected gain and class statistics of the candidate	after verification	publish the candidate or fall back
True partition ceilings, clean teacher statistics, test accuracy	offline only	explain losses and sensitivity (Opt., Table VI)

after the outcome, accepts 0.0% on Usenet and 77.4% on MNIST (Table IX): the verified target, not the verifier, decides whether Usenet publishes. No MNIST candidate meets the all-class recall target at this budget.

c) Auditing the ceiling is public naming: Estimating the one-to-one ceiling from a labelled audit finds the best assignment on its confusion matrix, so the estimate is the in-sample accuracy of the naming the audit induces. Figure 6 compares the two on audits from the development split; test labels only score. With one label per class, the CIFAR estimate is 92.5% while the induced naming achieves 62.2% on test; at five per class they meet at 91.6/91.6%, the ceiling. The others converge more slowly (MNIST and Usenet at 50 per class: 54.9/51.5% and 45.8/40.2%, one-to-one ceilings 52.6/42.4%). The estimate became accurate only once the induced naming neared the ceiling; the ceiling is an *ex post* diagnostic.

F. Natural writers and text clients

FEMNIST proxy purity rises from .361 at K=31 to .654 at K=500, but at 2k writers finer regions help only at $\varepsilon = 1$ (K=124: 38.63% versus 35.57% at K=62) and hurt at $\varepsilon = .1$ (K=31/62/124: $22.68/19.80/12.63\%$); 5000 public labels reach $50.65/59.33\%$ at 180/600 steps. Usenet's public portfolio matches or exceeds joint naming at every random-label budget (both 63.90% balanced at 300 labels).

VII. LESSONS FOR DATA PLATFORMS

Only the first two rows of Table X support decisions.

a) Buy: fix the acquisition contract before the privacy budget: Private queries pay where labels are only passively available; the per-draw distribution (median 13.6, fifth percentile -3.8) states what one deployment should expect.

b) Release: choose the interface by payload and partition fit: The regional report's advantage over pooled per-example votes is payload, not noise; a head pays where the partition bounds naming, and $v \approx \sqrt{TK/\rho_\varepsilon}/n$ predicts the noise term before any private data are touched.

c) Verify: budget evidence for the declared target: Mean gain is cheap but certifies no class. A small pilot chooses the split for class targets, and the calculator turns n, C and expected effects into a budget. The verified metric must be the reported one: a balanced-accuracy rule publishes no Usenet candidate.

d) Scope: Coverage is for the finite held-out cohort; tasks are benchmark-scale with frozen encoders (DINOv2, ResNet-18 and TF-IDF); image clients are simulated privacy units built from public benchmarks (with source-disjoint controls), Usenet authors and FEMNIST writers natural; tabular data are not covered. Zero-shot heads on other encoders change the representation resource and are not in $\mathcal{H}_{\text{pub}}$.

VIII. RELATED WORK

a) Differential privacy in data management: Differential privacy [12], [13] is built into query engines (PINQ [14], elastic-sensitivity SQL [15], PrivateSQL [16]), synthetic data [17] and benchmark methodology [18]. Federated databases add secure computation [19], [20]; local DP collects without a trusted curator [21], [22]. Here protected answers become supervision, and verification shares their budget.

b) Data acquisition, labelling and valuation: Labelling is a first-class pipeline cost [23], [24]: weak supervision creates labels [25], active learning chooses them [26], and valuation and marketplaces price data [27]–[29].

c) Private evaluation and certification: Confusion-matrix metrics [30], federated precision, recall and ROC-AUC [31], [32] and federated A/B tests [33] can be released privately. Private tuning and validation charge selection [11], [34]–[36], emit-or-abstain rules release a claim only when a private lower bound passes a threshold [37], [38], and simultaneous group bounds support non-private fairness audits [39]. Here the same budget also names the candidate.

d) Private learning, distillation and cluster reuse: DP-SGD and client-private federated averaging bound contributions [40]–[43]; secure aggregation with distributed discrete-Gaussian noise removes a trusted curator [1], [4], [5], DP-FTRL uses correlated noise [44]. Public features help private learning [45], [46]. Prediction exchange [47]–[51] and PATE-style teacher votes [52]–[54] transfer knowledge without parameters. NoiseCluster reuses a noisy cluster majority under label privacy ($\sqrt{2}$ sensitivity [55]); replacing a whole client gives $\sqrt{2K}$, so the two budgets are not interchangeable. Prio, Prochlo, split-and-mix and shuffling support aggregate-only interfaces [2], [3], [56]–[58], and browser reporting, summary releases and data markets bounded contributions [59]–[62].

IX. CONCLUSION

Label acquisition decides whether private supervision adds value; payload and partition fit, not raw sensitivity, separate release interfaces; and publication evidence is a separate purchase that a pilot population can price before deployment. Evaluations should report the acquisition contract, release payload and verified target alongside accuracy.

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